Check for updates

Transactions in GIS, 2008, 12(4): 475–491

Research Article

Agent-Based Simulation of Urban Residential Dynamics and Land Rent Change in a Gentrifying Area of Boston

Jeremy Jackson
Department of Geography
McGill University

Benjamin Forest
Department of Geography
McGill University

Raja Sengupta
Department of Geography
and McGill School of Environment
McGill University

Abstract
Certain complex processes are most effectively modeled not on the macro-scale, but from the bottom-up, by simulating the decisions of individual entities, or agents. This study uses an agent-based modeling (ABM) approach to simulate residential dynamics in an area of Boston that has increasingly experienced gentrification in the past decades. The model is instantiated using basic empirical data and uses simple decision-making rules, differentiated into four classes, to simulate the process of residential dynamics. The model employs the consumption explanation of the cause of gentrification, which emphasizes the choices of individuals drawn to urban amenities, while testing the production explanation, which suggests that major investments from the public and private sphere attract and explain gentrification. Verification shows that the processes in the model work according to its construction, simulates complexity and emergent phenomena, and may be a valuable explanatory tool for understanding and learning about some processes underlying gentrification.

1 Introduction

In many cities with strong creative and technological industries the presence of mobile young professionals and entrepreneurs drives up the real estate market, resulting in a phenomenon termed "gentrification". Although this process is not new and has been

Address for correspondence: Raja Sengupta, Department of Geography, 805 Sherbrooke Street West, McGill University, Montreal QC H3A2K6, Canada. E-mail: raja.sengupta@mcgill.ca

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd
doi: 10.1111/j.1467-9671.2008.01109.x

studied by urban geographers and planners since the 1970s (e.g. Smith 1979, Clay 1979, Lang 1982, Hamnett 1991) the return of young, wealthy professionals to the city and reinvestment in older urban cores has received renewed attention recently. In particular, Florida (2005) suggests that cities that do well in attracting skilled workers in the creative and technological industries (the so-called "creative class") will compete successfully in the globalizing economy (cf. Smith 2002). However, because this process also increases rents and home values in previously working-class city-center neighbourhoods, it displaces the non-professional class and elderly from these areas (Sassen 2001). Boston, with its thriving computer and biotechnology industries is an example of a global entrepreneurial city that has also experienced extensive gentrification in the last 20 years.

Gentrification is, however, a complex process, and it is often difficult to predict when and where this process will occur. Modeling complexity is best accomplished by "describing a system from the perspective of constituent units" (Bonabeau 2002, p. 7280), and by using a bottom-up approach (O'Sullivan 2002, Torrens and Nara 2007). The mechanisms of the process of gentrification are not explicit, and like with other complex processes the pattern "emerges" as a result of interactions of the underlying components and is different from the action of the individual components (Gilbert 2000). Portugali (2000), for instance, describes the "self-organizing" tendency of large groups of people in the city. Benenson et al.'s (2001) model describes how people order themselves within the city based on cultural, religious, or ethnic attributes in an ethnically segregated area of Tel Aviv. Models such as these emphasize interacting individual components that represent a bottom-up approach and fall under the general category of Agent-Based Models (ABMs), or more correctly Artificial Life Geospatial Agents (Sengupta and Sieber 2007).

ABMs mimic the perceived or measured behavioural response of an individual to an external stimuli using one of the many available computational models of boundedly rational decision-making behaviour as impacted by social networks (Sengupta and Sieber 2007), and can be used to represent individual people, businesses, or organizations that act (e.g. change location or farming practices) on certain rules and constraints. Such agents receive information about their local and global surroundings and use that information differently to make decisions about whether or not to move, where to move, and how to behave. Each agent "builds up its own representation of space and by acting, the agent transforms the space for others" (Benenson 1998). Interactions between agents and the landscape are central to this type of modeling. Thus, ABMs describe the complex interactions between entities and their environment inherent in natural processes better than other spatial modeling techniques (O'Sullivan 2008), and are therefore able to effectively model urban dynamics as a precursor to gentrification.

Here, we use ABMs to simulate the process of urban residential dynamics and land rent change in a gentrifying area, using a case study of several inner-city Boston, Massachusetts neighborhoods (Figure 1) to identify possible patterns of gentrification. The model uses four classes of agents ("professionals", "college students", "non-professionals", and "elderly"), and a set of simple decision rules, to predict changes in land values (i.e. assessed value of properties standardized by area) from 2003 to 2007. These classes were selected because they fit the demographic profile of the area and because they represent groups typically described in gentrification scenarios. The decision rules governing behaviours for each of these four classes of agents were obtained from a literature survey. Ethnicity was not considered in the model, but as will be discussed in the conclusions, does not seem to significantly affect the outcome in the area modeled. Further, investments in the area are observed to show no significant correlation with increases in land rents.

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

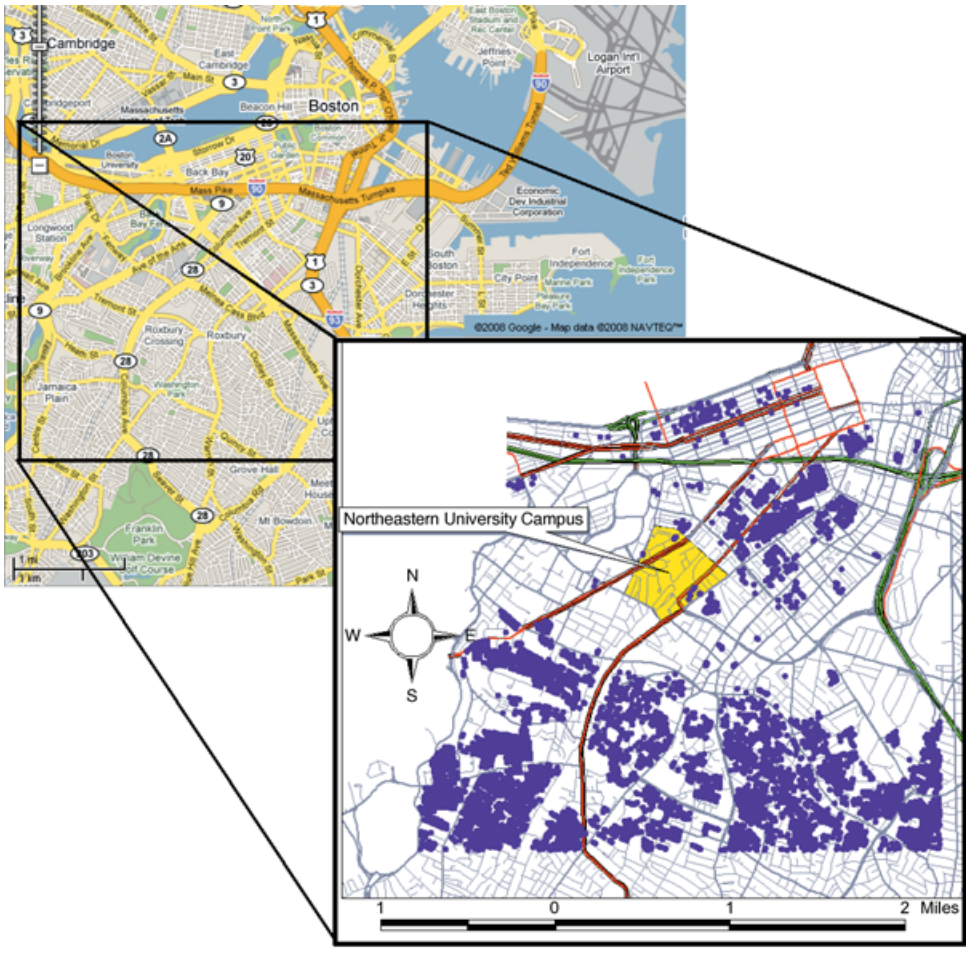


Figure 1 Map of the study area (surrounding Northeastern University). Blue dots represent the location of residential properties for which assessed values were available for 2003 and 2007

2 Model Design

The Recursive Porous Agent Simulation Toolkit (RePAST) is used to build and run this simulation (see http://repast.sourceforge.net for additional details). Within RePAST, the model uses a raster surface with information on each location's land rent and proximity to certain amenities to inform the decisions and movements of the four classes of agents who navigate the landscape using simple decision rules, which are based on the social class and economic ability of the agent. Agent movement is also influenced in some cases by memory of favorite locations, and for all agents by communication between friends of similar class.

A raster surface representing land rents (i.e. assessed value of properties in the Fenway/Kenmore and Roxbury neighborhoods, standardized to a "per square feet" value) in 2003 for the simulated region (Figure 1) was developed using Voronoi tesellation from a point file containing assessed building and land values received from the City of Boston Assessing Department. To this surface were added codes that represented distance from

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

desirable amenities (e.g. restaurants, pubs, stores and clubs), a university campus (i.e. Northeastern University), and commercial districts. This land rent surface was subsequently used by the agents within the RePAST simulation interface to guide their residential choices.

The design of the spatial information used in the simulation (i.e. the raster layer described above) was influenced in part by the nature of the agents' decision-making rules. Myers and Gearin (2001) show evidence that the general trend in the market, especially amongst wealthier populations is that there is an increased demand for dwellings in higher density areas and for homes that are closer to work and study locations. In addition, Ley's study (1986) showed evidence that a density of restaurants and art galleries shows a strong correlation with gentrification, establishing a foundation for broader arguments made by Florida (2002) and Clark (2004) that the growth of old, dense cities has been significantly driven by the preferences of young professionals for diversity and proximity to amenity-heavy locales. The evidence from all four studies mentioned above necessitates a decision-rule for professional agents that describe their preference to locate near dense commercial areas. In addition, the Myers and Gearin (2001) study confirms the contribution of an already logical decision-rule assigned for college students, i.e. they prefer to live close to the campus (which is their place of "work").

Agent classes are derived from Boston Housing Authority (2007) demographic data for the area and are divided as such: college student, professional, non-professional and elderly, and are assigned decision-rules accordingly. The number of non-professional and elderly agents in the simulation are static (reflecting the demographics of this area), whereas approximately 300 college students enter the simulation each year (presumably to attend Northeastern University). Since student life is transitory, approximately one in eight students migrate out of this class each year. Further, it is estimated that approximately 90% of college students join the professional workforce (Chapman and Gray 1999). A similar number of students therefore become "professional" agents each year. Similarly, because there has been an increase in the number of college graduates who receive non-college wages in recent years (Gottshalk and Hansen 1998, Chapman and Gray 1999), a certain percentage (10%) of college students also migrate into the non-professional class upon graduation. These transitions are a major part of the model's demographic dynamics, and are thought to contribute significantly to the process of gentrification (Smith 2005, Smith and Holt 2007).

Locational decisions of the various classes of agents are based on several factors: (1) the location is unoccupied by another agent; (2) that the location is close either to the college campus (for college students) or a commercial district (for professional agents); (3) the location's affordability; (4) the location has at least one neighbor of the same class; (5) agents are attracted to areas recommended by friends; and (6) some agents are attracted to areas they found amenable previously in the simulation. Preference for neighbor characteristics (factor 4) is based on a status/class rather than racial distinction, a technique used in other recent models on urban residential dynamics (Bernard and Willer 2007). Besides the factors of proximity to amenities, affordability and proximity to similar neighbors, two additional factors influence agent decisions: communication and memory. Communication of favoured locations between like-agents simulates social networks and further reinforces the attractiveness of certain locations. With spatial memory, the model assumes that college students are likely to continue to return to the areas they inhabited as students, provided their basic search criteria are met. This is based on the premise that students, as drivers of gentrification, often choose "favourite" locations to which they return in the future following their transition to the "professional" or "non-professional" class.

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

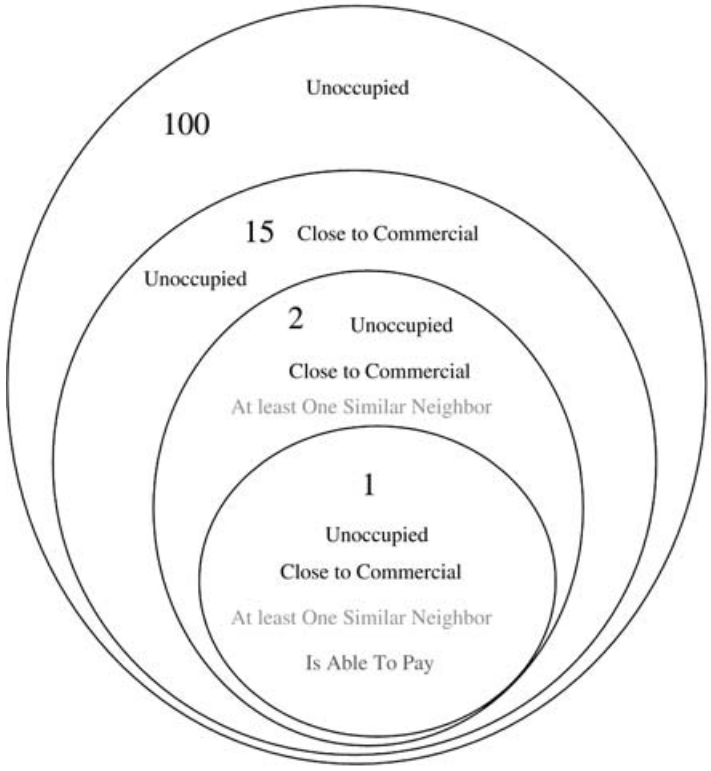


Figure 2 Average number and type of locations considered by a professional agent in its search for a new location. For example, of the 100 locations searched that are unoccupied, only 15 are also close to commercial districts. Of those that are both close to commercial districts and unoccupied, two are found to also have at least one neighboring location inhabited by an agent of similar class, and on average one out of two of those locations are affordable

The four factors described above, along with the influence of friendship networks and memory all combine to form the decision-making framework for the agents. For example, the typical steps taken by professional agents in searching for an "ideal" residential location is described by the diagram in Figure 2, and the decision rules are elaborated for each agent class in Figure 3.

Another issue is the question of when to move. Agent mobility is driven both by necessity and leisure. Benenson (1998) describes residential agent movement driven by local cognitive dissonance (based on social identity) and economic imperative. In our model, agent movement is driven by economic imperative in that an agent decides to look for a new location when it can no longer afford its current place. Cost was found by Clark (1983) to be the second most common reason for household mobility behind changes in household structure, thus it is an appropriate representation of the choices of residents. Because it is an explanatory model and movement must be maximized to observe the patterns of geospatial activity, agents are also programmed to move regularly even if not caused to do so by economic factors. This is similar to techniques used in other such research, including some that has simulated movement of residential agents at regular intervals (Benenson 1998). The movement rules, as described here, are

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

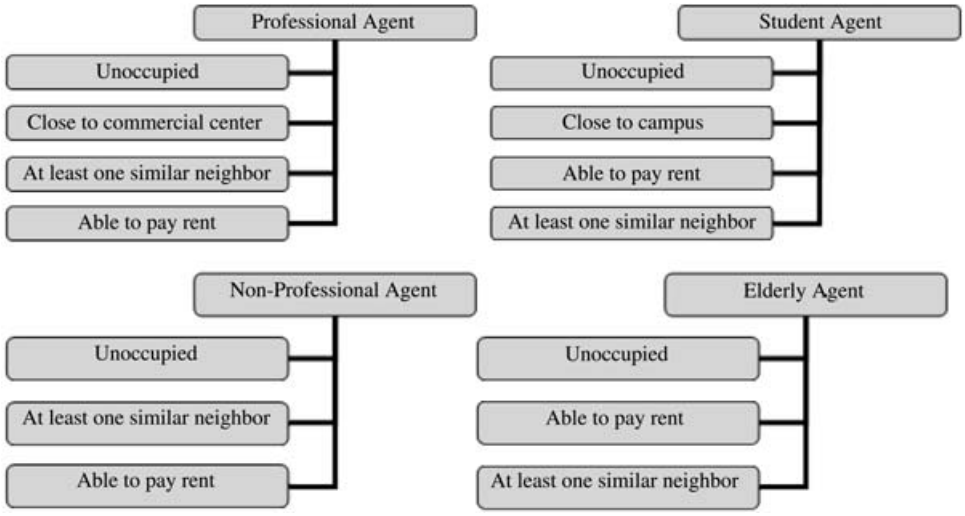


Figure 3 Decision rules for each agent class, with the rules at the top given the highest priority during a search for a location, and those at the bottom given the lowest

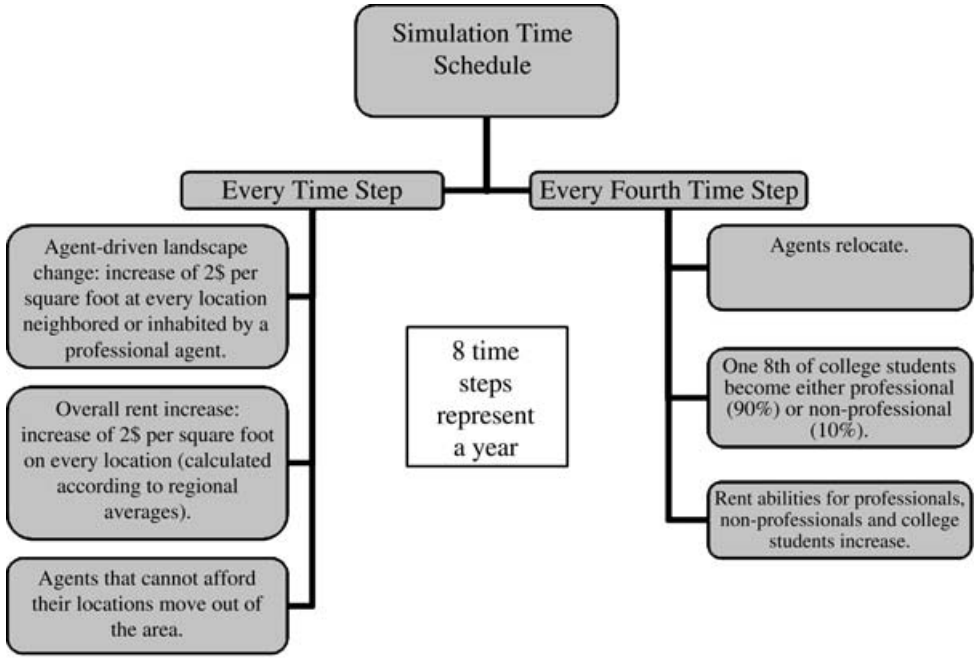


Figure 4 Schedule of execution of events in RePAST

implemented according to the model's schedule and parameterization which are described below.

Figure 4 shows the development of the schedule for the events that comprise the simulation. At every time step, the rents of every location on the land rent surface are

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

increased by $2 per square foot, an estimation derived from data from a regional housing study (Allen et al. 2002). The professional agents are programmed to cause an increase in the land rent of their properties and those of their neighbors at the same rate. Finally, it is noted that at the end of every time step agents who cannot afford to live at their current x-y coordinate are removed from the simulation. Between this interval and every fourth time step, agents of all classes except for the elderly increase their ability to pay higher rents. Initially, agents are assigned random locations that may or may not meet all their preferred requirements. Beginning at the fourth time step and every subsequent fourth time step, these agents relocate according to the rules described above. The demographic transition described earlier is also shown in Figure 4.

Having examined relocation schedules as well as location decisions, it is also helpful to examine the initial parameters of the simulation to better understand how the model is instantiated. The model displays five parameters that can be altered at the beginning of each run of the simulation: the number of agents for each class, the rent abilities for each class, the amount of energy an agent is willing to put into the search for new housing, the number of friends one has, and whether or not the agent uses information about prior residence in an area (e.g. memory). These were selected for the purpose of verifying the simulated processes. The rent abilities for each type of agent are set relative to one another based on data taken from the census and from an independent research organization (Allen et al. 2002). These data showed that the rent paid by professionals is approximately twice the amount paid by the non-professional class, and greater than twice that paid by the elderly (Allen et al. 2002). No statistics to determine the relative rent abilities of college students were found. It was reasoned, however, that as upwardly mobile individuals with some disposable income, college students can spend more on rent than the average working class household. In our simulation, the average land rent at the beginning of the simulation is $171 per square foot. Thus, the initial rent abilities are distributed as shown in Table 1.

The number of agents used in the simulation is another major parameter. Tests showed little difference in simulations using more than 4,000 agents, and increasing the number much beyond this slowed the computations considerably. Thus, the number of agents being modeled represents a proportion of the inhabitants of the area, with the proportion in each category determined from data obtained from the report of a city government agency (Boston Housing Authority 2007). At the start of the simulation, as shown in Table 1, there are 1,184 college, 1,654 professional, 972 non-professional and 196 elderly agents in the simulated area. The search energy is initialized as 100, meaning that agents will search a maximum of 100 locations before deciding to stop searching. The number of friends of each agent is initialized as five, all of which are in the same class,

Table 1 The initial number of agents for each class and their starting rent abilities

Agent Class	Number of Agents	Starting Rent Ability ($ per square ft)
Professional	1,654	145
College Student	1,184	75
Non-Professional	972	95
Elderly	196	65

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and are chosen randomly regardless of location. Finally, in most runs of the simulation, unless explained otherwise in the results section, memory is included as a factor.

Evaluation of the model is fairly straightforward. The correctness of the model's build is verified through sensitivity analysis procedures described in the following section, with the aim of verifying that the model captures complex and emergent patterns in the system. Then the differences in land rents between 2003 and 2007 from the city of Boston are compared against the spatial patterns of agent settlement to determine whether the model is able to capture processes of urban residential dynamics and land rent change. Finally, a test is performed to determine whether specific development investments further explain increases in land rents between 2003 and 2007.

3 Results

3.1 Simulation Results and Verification of the Model's Processes

To verify certain processes in the model, multiple outputs of the simulation with minor differences in the model's initial parameters were analyzed. The effects of demographic changes in the system were observed through the creation of a measurement tool in RePAST. Figure 5 shows population dynamics for the control run, where the simulation's initial parameters are not adjusted. In this simulation, the demographic profile responds the way it is expected to: agents gradually leave the simulation because of an inability to find affordable locations – this can be seen in the decrease in non-professionals, elderly, and college students. While the number of professional agents decreases initially, the population quickly rebounds and grows as college students transition into the professional agent phase. Additional tests that confirm the complex nature of the agent

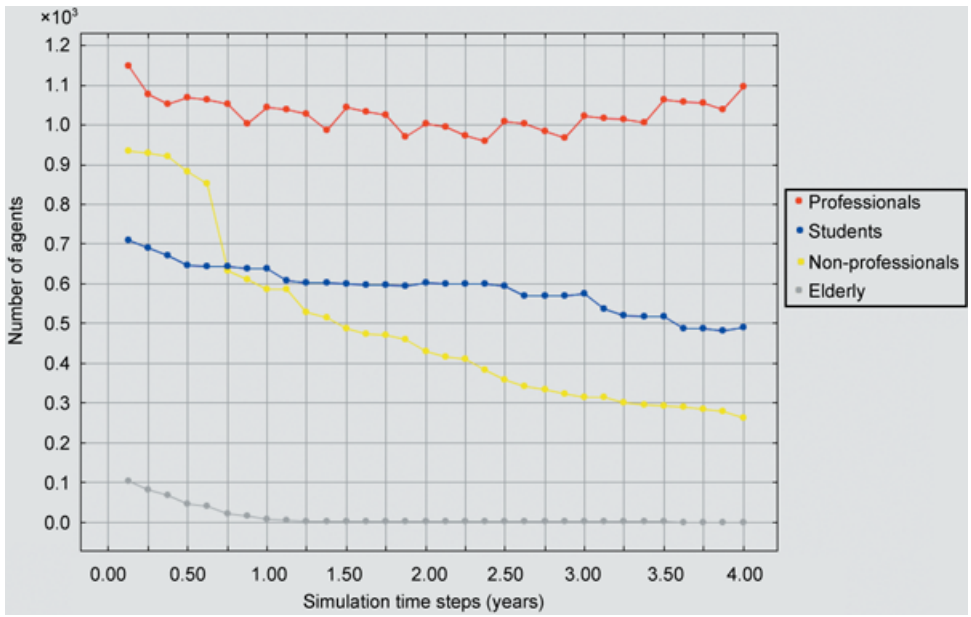


Figure 5 Trend of agent populations in the simulation over time for the four classes

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

simulation are presented below in Table 2 (with notes indicating whether the results were expected or not, as an indicator of complexity).

Verification of the effects of changes in rent abilities is one such test, and shows that in most cases the model performs as expected. Tests A through G in Table 2 examine the effects of small changes in agents' rent abilities and generally verify expected processes, finding that all variations in these initial settings produce expected demographic processes. From the way the model was constructed, one could hypothesize that should the rent ability of professionals decline, their populations would decrease somewhat as well. This is in fact the case. When professional agents' rent ability is decreased by 10%, the professional population decreases as expected (as seen in Test B in Table 2). When the professional agents' rent abilities are increased by 10%, the professional population stays well above levels reached in the control (Test A). Test D tests the interconnections between classes in the model: when the rent ability of college students is decreased by 10%, the college student category decreases by 35%. Additionally, the average professional population also decreases by 10%. This shows that the model successfully captures the area's demographic transition, where students who graduate become professionals and decide to stay in the area. Test E reveals some important information about the model – a minor increase in the non-professional's rent abilities causes a disproportionate (44%) increase in their average population, implying that the rent ability of the non-professional agent is at a tipping point. Tipping points occur when the conditions of a system are brought to a point where minor changes would cause a dramatic increase in the occurrence of rare phenomena (Gladwell 2000). This also offers evidence that the model captures complex system dynamics, where changes in rent abilities are not directly correlated with a set percent increase or decrease in the various types of agents. Finally, a basic test was performed to confirm that the model's depiction of the influence of professionals on land rent dynamics works as intended. Test M shows that when the code which enacts agent-driven landscape change was made inactive the rents in the landscape decreased on average by 2%, suggesting that the rents in the landscape are being influenced upwards by the professional agents as intended.

Besides testing the effects of changes in initial rent abilities, other tests were compiled to show whether the processes expressed in the model are the processes designed into the system. Test H, for instance, shows that as the search energy allotted to each agent increases by 50%, the average agent populations increase substantially. This is an expected outcome, given the structure of the model, because the more places an agent can examine to find a suitable location, the more likely that agent will eventually find one, and the less likely the agent will be removed from the area due to an inability to pay. While most tests shown in Table 2 exhibited expected outcomes, verification tests of a theoretical nature were also performed and are discussed below.

The model allows the user the option of making "memory" active or inactive, thereby affecting the tendency of an agent to return to a favorite location. Memory was used by Bennett and Tang (2005) within ABMs to simulate the return of migrating Elk to certain locations, and its use here is analogous. In addition, because communication within friendship networks is considered to be an important input to the residential decision-making process, their effects on the outcome of the proposed model are considered. Each agent initially has five friends with whom the agent can communicate information (e.g. message passing) throughout the simulation. By setting this number to zero, the user effectively makes the message passing functionality in the system inactive.

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2 Tests carried out for verifying the processes simulated in the model, and evaluation of the outcome

Verification Test	Result	Expected Outcome?
A. Professional rent ability decreased by 10%	Professional population lower on average by 10%	Yes
B. Professional rent ability increased by 10%	Professional population higher on average by 13%	Yes
C. College student rent ability increased by 10%	College student population higher on average by 13%	Yes
D. College student rent ability decreased by 10%	College student population lower on average by 35%	Yes
	Professional population lower on average by 10%	No
E. Non-professional rent ability increased by 10%	Non-professional population higher on average by 44%	Yes
F. Non-professional rent ability decreased by 10%	Non-professional population lower on average by 11%	Yes
G. Elderly agent rent ability increased by 10%	Elderly population higher on average by 10%	Yes
H. Search energy increased by 50%	Professional population higher on average by 8%	Yes
	College student population higher on average by 5%	Yes
	Non-professional population higher on average by 10%	Yes
I. Initial professional population increased by 25%	No significant impact on rents or on other populations	No
J. Initial professional population increased by 25% and rent ability by 50%	Average land rent across landscape throughout simulation decreases by 1%	No – expected more
	Non-professional agent population decreases by 12%	Yes
K. Spatial memory made inactive	Professional population higher on average by 14%	No
	Non-professional population higher on average by 60%	No
L. Communication made inactive	Professional population higher on average by 10%	No
	Non-professional population higher on average by 37%	No
	Less clustering observed	Yes
M. Agent-driven landscape change made inactive	Average land rent across landscape throughout simulation decreases by 2%	Yes

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2020]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

In this model, message passing through friendship provides a professional agent with the ability to draw in another professional living outside of a commercial district and bring him/her to a location that may otherwise be attractive only because of its affordable rent. Thus, when a professional agent is "informed" that a friend has found a good location, it will choose to locate near the friend. Memory will also play a secondary role in this choice, as the agent will try to locate itself both near a friend as well as in a previously inhabited area. Such message passing occurs not only for the professional agent class, but the case of the professional is highlighted because of its potential impact on the land rent environment according to the model's construction.

Tests K and L show the effects on the model when memory and message passing are made inactive. In both cases, no changes in rents or populations were expected, only that fewer agents would stay in one location for the duration of the simulation, but results showed that populations of both professionals and non-professionals are affected when spatial memory is made inactive. Observations indicate that with message passing made active, several large clusters of professionals are formed. Conversely, when message passing is removed from the system, the professional agents are not as clustered, thus gentrification occurs in more areas. This result was expected. However, in tests K and L in Table 2, when communication between agents is made inactive, both professional and non-professional populations are observed to increase. This effect was not expected. One explanation may be because of the structure of the decision-making process: a professional agent that is placed on the property a friend recommends will likely stay there until the rents increase and it can no longer afford that location. These agents are subsequently removed from the simulation. On the other hand, when agents are not given any suggestions of locations from friends, they then choose randomly from the options available, and find locations that are affordable and move when they can no longer afford that location. Thus, these tests expose the need in the model to incorporate a greater diversity of reasons for relocation by agents.

In summary, most analysis suggests that the model displays processes according to its design; however, there are situations when the model behaviour deviates from expected to produce emergent behaviour. Minor changes in the initial rent abilities of agents show expected changes in populations of their respective classes, some also causing changes in populations of other classes, this being a possible indicator of complexity. Rent ability tests also show that the model may have uncovered a tipping point for the rent ability of non-professional which if attained, can cause a significant increase in their populations in the given simulation. Tests of the effects of increasing the search energy of agents and making agent-driven landscape change functions inactive verified that the model demonstrates the processes it was intended to. Other tests offered a mix of surprises, and lessons for future models. When spatial memory and communication are made inactive, results are not as expected, though reasons for these surprises have been explained. The most persuasive evidence that the model captures complexity in the system is that when the initial rent ability and population of professional agents are increased, the population of non-professionals is significantly impacted (decreased), because of competition with the professional agents. Having shown that the model works according to specification, it is now appropriate to evaluate the simulation results for its verisimilitude to realistic outcomes. Model results are examined in the following sections by analyzing predictions of where gentrification will occur vis-à-vis actual increases in land rents between 2003–2007, and whether the incorporation of major investments in the area could improve the model's accuracy.

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3.2 Areas of Gentrification

Two factors are believed to be an effective measure of gentrification. The first is displacement, meaning a replacement of non-professionals, working-class or poor residents with a wealthier class, usually professionals; the process is indicated in demographic analysis by a relative increase in the presence of the latter, termed as "gentrifiers" (Atkinson 2000), and a decrease in the presence of the former, i.e. the displaced. Another sign that an area is gentrifying is that it experiences an increase in property values of the area relative to those of the region as a whole (Galster and Peacock 1986). Observations of increase in land rents between 2003–2007 compared to the final location of agents in the simulation, as shown in Figure 6, suggest that these processes may be occurring in the regions circled. In running the simulation, it is possible to observe the demographic transition and subsequent gentrification, as agents in these areas start out as college students looking for affordable rents in a historically working-class/poor neighborhood (Roxbury), and then become professionals and return to live in these areas. The process of college students becoming professionals, it has been argued, is a common present form of gentrification (Smith 2005, 2007). However, it is difficult to draw conclusive results or make any predictions about gentrification in specific areas.

3.3 Testing the Theory of Structural Investment

There is a structural (production) perspective on gentrification that suggests that investments are the driving force behind an increase in land values (Smith 1979). As investment is not accounted for in the agent-based model (which follows more of an individual/consumption perspective), it was hypothesized that there would be an association between the location of actual investments and areas experiencing an increase in land rents. To test this structural investment perspective, locations of large investment projects that were either constructed or under construction from 2003 to 2007 were obtained from the Boston Redevelopment Authority. Addresses were geocoded, and land rent increases between 2003–2007 within 800 m (the distance of roughly two street blocks in this area) from these locations were evaluated using buffers.

Results indicate that areas showing increases in land rent were not significantly associated with the locations of investment. A t-test showed no statistical distinction between the mean value of the difference between predicted and observed land values for cells within 800 ft of investments ($27.7/sq.ft.), and cells further away ($27.2/sq.ft.). In short, knowing the location of investments does not allow one to predict changes in land value any better than the agent-based model already does. This supports the individual/consumption perspective on gentrification, and suggests that the factors accounted for in our model may be sufficient for this case study.

4 Discussion

The Agent-Based Model (ABM) described above uses simple decision rules and selection criteria, yet manages to capture important elements of gentrification and demonstrates the potential for the use of ABM as a comprehensive analytical tool for this complex social process. However, to evaluate results, it is important to discuss model limitations in detail.

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

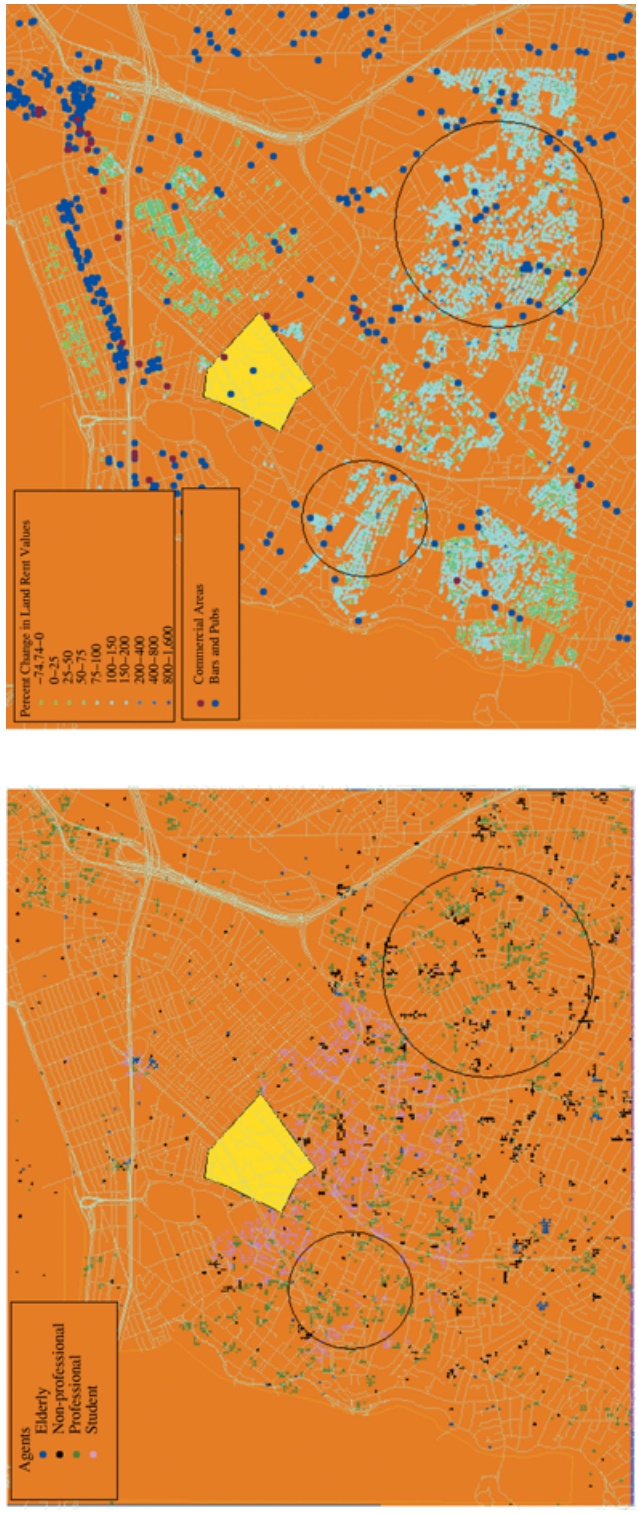


Figure 6 Maps showing (a) locational choices of professional agents; and (b) the percent change in land rent values between 2003–2007 (areas showing the highest percentage increase in reality are circled in both maps)

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

4.1 How Many Rules, How Many Variables?

Rules chosen for this model are simple, in part because of data limitations, but also by design. The aim of this research was not to capture the specific choices of each individual, but to capture broad spatial patterns that give rise to gentrification. The demographic transition modeled in our research better reflects the lifecycle transition experienced in the city than would traditional household lifecycle models. Regardless, the relevance of such lifecycle models for housing mobility is still a subject of debate (Clark 1983), and should not be considered essential for the analysis. In our model, most college students become professionals and stay in the area. When considering locations, professionals decide to move to areas where they once lived because research shows that where students decide to live, gentrification often follows (Smith 2005, 2007). Other explicit measurements of housing dissatisfaction such as housing space, quality, and public services of the neighborhood, are not considered for lack of available data. Further, note that in Torrens and Nara's (2007, p. 344) model (which accounts for housing choices of individuals), there are many more variables considered when compared to the model presented for this research; on the regional scale these include median property value, housing type, median economic status, median property size median accessibility and the ethnic profile, while the agent also considers these same variables for the individual house or home. Incorporating more variables such as these, and those argued by Clark (1983) to influence mobility would be the next step for the model's development. However, the rules chosen for our model seem to capture at least some essential elements of complexity in the system being modeled.

4.2 Scale

The model incorporates three important levels of organization described in the literature as essential to capturing the complexity of urban systems: the individual, the local and the global. The local level refers to the states of the neighbors, and the global refers to the characteristics of the whole city (Portugali et al. 1997, Benenson 1998). By comparison, the model presented herein utilizes all three levels – at the individual level, it looks at the choices of theoretical residents with specific economic and class distinctiveness; at the local level it predicts that the locational choices of these individuals will be influenced by the potential characteristics of neighboring residents; and at the global level the rent increases simulated for the whole surface are derived from city-wide data, thus the movements of agents are impacted by known global economic factors. The model thus makes a contribution to the field in that it offers a relatively simple simulation framework that involves all the essential levels of urban organization and could be applicable for many case studies were certain data available.

4.3 Race and Ethnicity

Research on urban residential segregation indicates race and ethnicity as playing a major role in the location choices of individuals (Fossett 2006; Clark 1986, 1992). Indeed, seminal work on modelling segregation by Schelling (1969), and Benenson et al.'s (2001) work using agent-based models of residential location choice cite race and ethnicity as key factors controlling patterns of settlement. However, the 2000 census block group data for the modeled area suggested that the largest increases in land rent values have

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

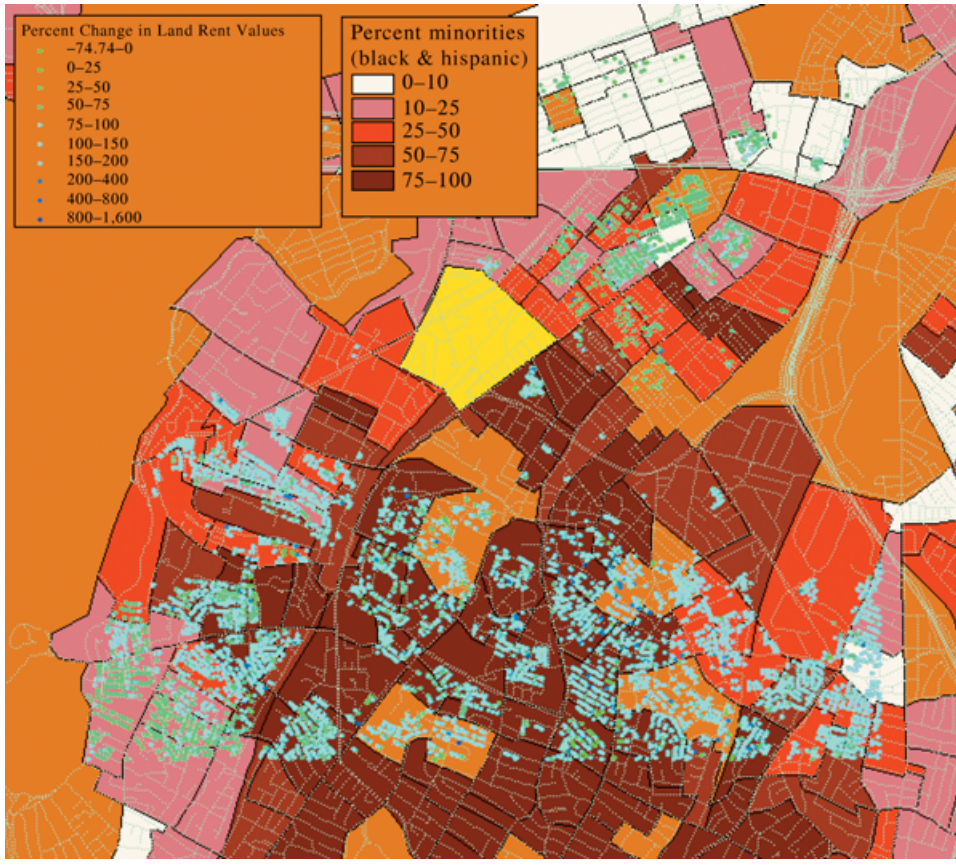


Figure 7 Percent change in land rent values between 2003–2007 as compared to percent minorities (blacks and hispanics) by block group (2000 census)

occurred in minority-dominated areas (i.e. block groups where Hispanic or Black populations are the majority) (Figure 7), suggesting that gentrifiers prefer racially diverse neighbourhoods. This is contrary to the expectation that residential location choice is highly influenced by racial prejudice and suggests, as has already argued in other urban contexts (Bruch and Mare 2006) that decision-making with respect to residential choice may be more nuanced and not significantly influenced by race. For this reason, the model uses economic status instead of race during simulation. In other words, class is the most significant factor in determining the categorization and thus eventual location of individuals.

5 Conclusions

The agent-based modeling (ABM) approach presented herein uses basic empirical data and simple decision-making rules to simulate residential dynamics. Further, it breaks demographics down by class, not by race, assigning simple preference rules for each residential agent. Simplified decision rules and lack of racial preferences notwithstanding,

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

verification of the model's processes shows that the model construction is reliable and a good start at capturing the complexity and emergent nature of urban residential dynamics. The identification of a "tipping point" based on the rent ability of the "non-professional" class, as well as the indirect impact of the rent abilities of the professional class on populations of other classes, can be seen as indicators of complexity in the system. Further, consideration of infrastructure investments appears to add little value to explaining increase in land rent values. While the effectiveness of such a model in predicting gentrification is difficult to validate, it is argued that this model of individual decision-making and agent-environment interaction is a valuable contribution towards improving the understanding of some of the processes underlying gentrification dynamics in global, growing cities.

References

Allen R, Heudorfer B, and Weismann G 2002 *The Greater Boston Housing Report Card*. Boston, MA, Northeastern University Center for Urban and Regional Planning

Atkinson R 2000 Professionalization and displacement in Greater London. *Area* 32: 287–95

Benenson I 1998 Multi-agent simulations of residential dynamics in the city. *Computers, Environment and Urban Systems* 22: 25–42

Benenson I, Omer I, and Hatna E 2001 Entity-based modeling of urban residential dynamics: The case of Yaffo Tel Aviv. *Environment and Planning B* 29: 491–512

Bennett D A and Tang W 2006 Modelling adaptive, spatially aware, and mobile agents: Elk migration in Yellowstone. *International Journal of Geographical Information Science* 20: 1039–66

Bernard S and Willer R 2007 A wealth and status-based model of residential segregation. *Journal of Mathematical Sociology* 31: 149–74

Bonabeau E 2002 Agent-based modelling: Methods and techniques for simulating human systems. *Proceedings of the National Academy of Science* 99: 7280–7

Boston Housing Authority 2007 *Fenway-Kenmore 2000 Census of Population and Housing*. Boston, MA, City of Boston Report No 577

Bruch E and Mare R 2006 Neighborhood choice and neighborhood change. *American Journal of Sociology* 112: 667–709

Chapman R and Gray G 1999 Conflicting signals: The labor market and college educated workers. *Journal of Economic Issues* 33: 661–76

Clark W 1983 Life cycle and housing adjustment as explanations of residential mobility. *Urban Studies* 20: 47–57

Clark W 1986 Residential segregation in American cities: A review and interpretation. *Population Research and Policy Review* 5: 95–127

Clark W 1992 Residential preferences and residential choices in a multiethnic context. *Demography* 29: 451–66

Clark T N 2004 The city as an entertainment machine. Oxford, Elsevier

Clay P 1979 *Neighborhood Renewal: Middle Class Resettlement and Incumbent Upgrading in American Neighborhoods*. Lexington, MA, Lexington Books

Florida R L 2002 *The Rise of the Creative Class*. New York, Basic Books

Florida R L 2005 *The Flight of the Creative Class: The New Global Competition for Talent*. New York, Harper Business

Fossett M 2006 Ethnic preferences, social distance dynamics, and residential segregation: Theoretical explorations using simulation analysis. *Journal of Mathematical Sociology* 30: 185–274

Galster G and Peacock S 1986 Urban gentrification: Evaluating alternative indicators. *Social Indicators Research* 18: 321–37

Gilbert N T 2000 How to build and use agent-based models in social science. *Mind and Society* 1: 57–72

Gladwell M 2000 *Tipping Points: How Little Things Can Make a Big Difference*. Boston, MA, Little, Brown and Company

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gottshalk P and Hansen M 1998 *Have Recent College Graduates Experienced Worsening Wage and Job Distributions?* Boston, MA, Boston College Department of Economics Working Papers

Hamnett C 1991 The blind men and the elephant: The explanation of gentrification. *Transactions of the Institute of British Geographers* 16: 173–89

Lang M E 1982 *Gentrification and Urban Decline: Strategies for America's Older Cities*. Cambridge, MA, Ballinger Press

Ley D 1986 Alternative explanations for inner-city gentrification: A Canadian assessment. *Annals of the Association of American Geographers* 76: 521–35

Myers D and Gearin E 2001 Current preferences and future demand for denser residential environments. *Housing Policy Debate* 12: 633–59

O'Sullivan D 2002 Toward micro-scale spatial modeling of gentrification. *Journal of Geographical Systems* 4: 251–74

O'Sullivan D 2008 Geographical information science: Agent-based models. *Progress in Human Geography* 10: 1–10

Portugali J 2000 *Self-Organization and the City*. New York, Springer

Portugali J, Benenson I, and Omer I 1997 Spatial cognitive dissonance and socio-spatial emergence in a self-organizing city. *Environment and Planning B* 24: 263–85

Sassen S 2001 *The Global City: New York, London, Tokyo*. Princeton, NJ, Princeton University Press

Schelling T C 1969 Models of segregation. *American Economic Review* 59: 488–93

Sengupta R and Sieber S 2007 Geospatial agents, agents everywhere . . . *Transactions in GIS* 11: 483–506

Smith N 1979 Toward a theory of gentrification: A back to the city movement by capital, not people. *Journal of the American Planning Association* 45: 538–48

Smith N 2002 New globalism, new urbanism: Gentrification as global urban strategy *Antipode* 34: 427–50

Smith D P 2005 'Studentification': The gentrification factory? In Atkinson R G B (ed) *Gentrification in a Global Context: The New Urban Colonialism*. London, Routledge: 72–89

Smith D P and Holt L 2007 Studentification and 'apprentice' gentrifiers within Britain's provincial towns and cities: Extending the meaning of gentrification. *Environment and Planning A* 39: 142–61

Torrens P and Nara A 2007 Modeling gentrification dynamics: A hybrid approach. *Computers, Environment and Urban Systems* 31: 337–61

© 2008 The Authors. Journal compilation © 2008 Blackwell Publishing Ltd

14679671, 2008, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1467-9671.2008.01109.x by The University Of Melbourne, Wiley Online Library on [04/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License